

» Baolai Steel



Be a leading steel pipe manufacturer
& piping systematic solution supplier

01 COMPANY PROFILE

天津宝来钢铁集团

天津宝来钢铁集团是一家以无缝钢管、焊接钢管及其他钢材产品为主，集工、贸、商于一体的综合性企业。宝来成立于 1991 年，是国内最早开发和生产普通和特殊直径钢管的少数几家专业公司之一。

公司连续两届被评为“天津市百强企业”，连续五届跻身“中国制造业 500 强”，银行资信等级为“AAA”级，并通过了 ISO、API、FM、UL、CE、JIS、QA、QM 等认证。

天津宝来富特国际贸易有限公司是天津宝来钢铁集团的全资子公司，全面负责各类钢材产品的全球营销和销售。

主要出口产品有：消防钢管、焊接钢管、无缝钢管、直缝埋弧焊钢管、埋弧焊钢管、钢卷、钢板和钢结构产品，以及各类管件、泵和阀门。

我们还提供镀锌、FBE、3PE、3LPP 和管道深加工等定制服务。此外，我们还有一支专门的团队提供技术支持，就钢材的正确使用提供建议，并协助客户以最佳方式使用产品。

宝来拥有约 1300 名员工、13 条生产线、多条加工线和 4 个拥有先进技术的实验室。我们的年生产能力约为 500 万吨，每年出口 15 万吨到南美、北美、欧洲、中东、非洲、澳大利亚和东南亚。

我们热忱欢迎和期待长期、新的和潜在的宝贵客户和顾客的兴趣、咨询和订单，并期待现在和将来成为您管材和其他产品的首选来源和长期合作伙伴。



TIANJIN BAOLAI STEEL GROUP is a comprehensive enterprise with industry, trade and commerce, which focus on seamless pipe, welded steel pipes and other steel products. Established in 1991, Baolai is among the first few specialized companies that develop and manufacture ordinary and special diameter steel pipes.

We have been awarded "Top 100 Enterprises of Tianjin" for two consecutive years, titled among the "TOP 500 Enterprises in China Manufacturing Industry" for 5 consecutive years and a credit rating "AAA" by Banks and have obtained certifications, ISO, API, FM, UL, CE, JIS, QA & QM, etc.

Tianjin Baolai Fortune Trade Co., Ltd is a wholly owned subsidiary of TIANJIN BAOLAI STEEL GROUP, fully responsible for global marketing and sales of all types of steel products.

The principal products for export are: welded steel pipes, seamless steel pipes, fire-fighting steel pipes, LSAW steel pipes, SSAW steel pipes, steel coils, steel sheets and steel structure products, as well as various types of pipe fittings, pumps and valves.

We also offer customized services in galvanized, FBE, 3PE, 3LPP and deep processing of pipes. Furthermore, a dedicated team is available to provide technical support in order to advise on the correct use of steel and to assist customers in using the products to their best advantage.

Baolai have about 1300 employees and 13 production lines, many processing lines and 4 laboratories with advanced technology now. Our annual capacity is approximately 5 million tons and exports of 180,000 tons per year to South and North America, Europe, the Middle East, Africa, Australia and Southeast Asia.

We warmly welcome and anticipate interest, inquiries and orders from the valuable customers and clients, the long time, new and potential ones, and look forward to becoming your source of first choice and long term partner for pipe and tube and other products now and in the future.





ENTERPRISE QUALIFICATION



MAIN STANDARD AND PRODUCTS

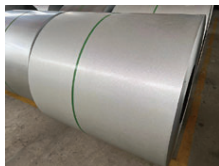
Main Standard

Standard	Standard Name	Grade	Outside Diameter	Wall Thickness
API 5L	Pipeline Steel Pipe Specification	GR.B, X42, X46, 52, X56, X60, X65, X70, L320, L360, L390, L415	1/8"-42"	SCH10-SCHXXS
API 5CT	Casing And Tubing	H40,J55,K55,N80,L80,C90,R95,T95,P110,C110,Q125	4-1/2"-20"	5.21 - 16.13 mm
ASTM A53	Pipe, Steel, Black And Hot-Dipped, Zinc-Coated, Welded And Seamless	Gr.A,Gr.B	1/8"-42"	SCH10-SCHXXS
ASTM A106	Steel Carbon Steel Pipe For High-Temperature Service	Gr.A Gr.B Gr.C	1/8"-42"	SCH10-SCHXXS
ASTM A500	Cold Formed Welded And Seamless Carbon Steel Structural Tubing In Round And Shapes	Gr.A Gr.B Gr.C Gr.D	10*10-500*500	1.7-15.75
ASTM A795	Black And Hot Dipped Zinc Coated (Galvanized) Welded And Seamless Steel Pipe For Protection Use	Gr.A Gr.B	3/4"-10"	1.7-13.75
ASTM A179	Cold-Drawn Seamless Mild Steel Tubes For Heat-Exchanger And Condenser Tubes		1/4"---4 3/4"	0.02" - 0.375"
ASTM A192	Seamless Carbon Steel Boiler Tubes For High Pressure Service		1/4"---4 3/4"	0.02" - 0.375"
ASTM A252	Welded And Seamless Steel Pipe Piles	Gr.1 Gr.2 Gr.3	6" - 24"	0.134" - -- 0.5"
JIS G3452	Carbon steel pipe for ordinary piping	Gr.SGP	1/2"-10"	1.7-15.75
EN 10210	Hot finished structural hollow section for non-alloy and fine grain steels	S235JRH, S275J0H, S275J2H, S355J0H, S355J2H, S355K2H	20*30-1000*800mm	1.3-15.75

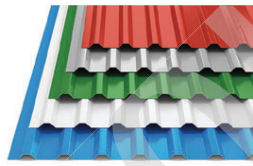
Main Products



PPGI Coils



Hot-Dipped Galvalume Steel Coil



Steel Profile Sheet



Hot Rolled Steel Coil



Fire System Steel Pipe



Fire Pipe Fittings



Carbon Pipe Fittings



ERW Steel Pipe



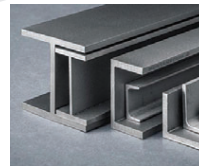
Solar Mounting System



Steel Prop Jack



Scaffolding



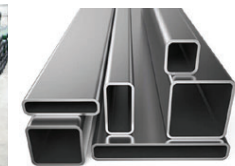
Section Steel



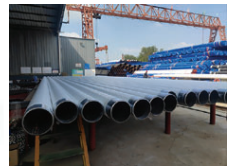
Seamless Black Steel Pipe



OCTG



Hollow Section Steel Pipe



3PP Coating Pipe

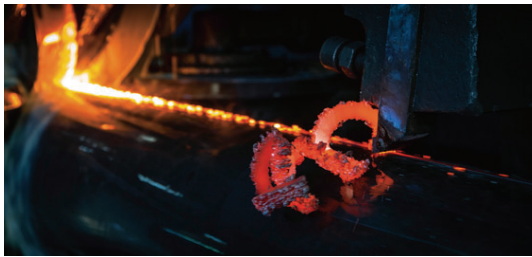
02 PRODUCT DISPLAY

1 WELDED STEEL PIPE

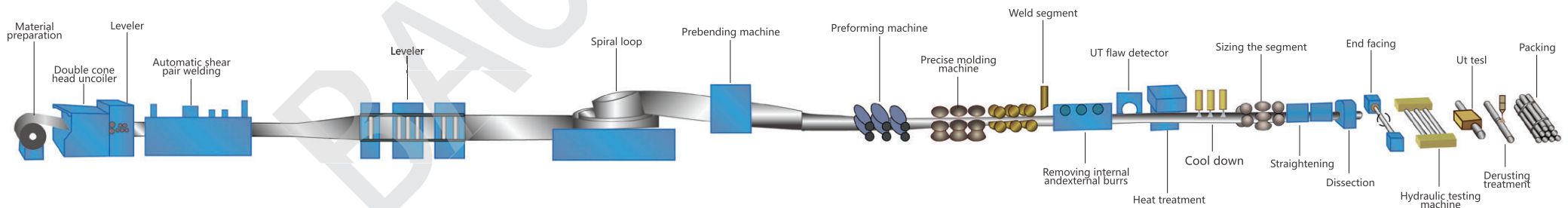
Welded steel pipe is a seamed steel pipe. The production process is to bend the pipe blank (steel plate and steel strip) into a tube of the required cross-sectional shape and size by various forming methods, and then weld the weld by different welding methods. The process of obtaining a steel

pipe. Compared with seamless steel pipes, welded pipes have the characteristics of high product precision, especially wall thickness accuracy, simple main equipment, small footprint, continuous operation in production, flexible production, and wide product range of the unit.

From the production process, welded pipes are further divided into: ERW (Electric Resistance Welding); LSAW (Longitudinally Submerged Arc Welding); SSAW (Spiral Submerged Arc Welding); Hollow Sections; EMT & IMC Pipe.



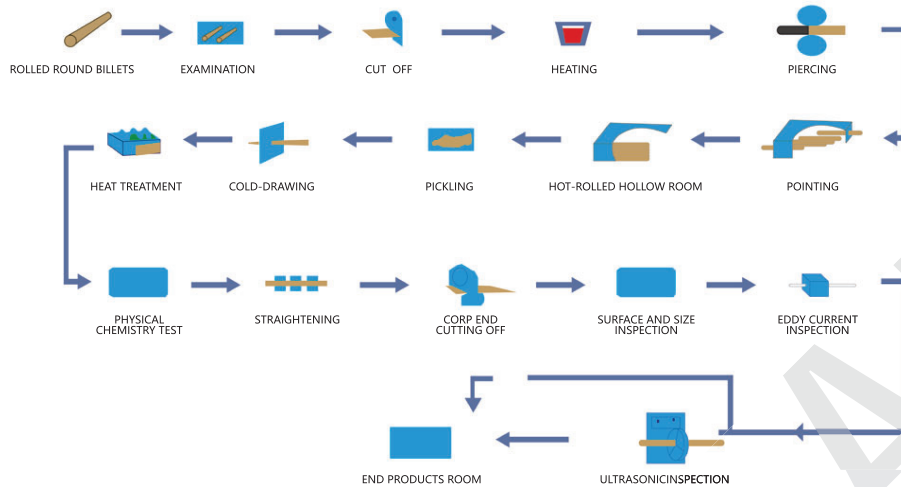
Manufacturing Processes



2 SEAMLESS STEEL PIPE

Seamless steel pipes are perforated from whole round steel, and steel pipes without welds on the surface are called seamless steel pipes. According to the production method, seamless steel pipes can be divided into hot-rolled seamless steel pipes, cold-rolled seamless steel pipes, cold-drawn seamless steel pipes, extruded seamless steel pipes, and top pipes.

COLD DRAWN SEAMLESS STEEL PIPE process flow chart



Cold Drawn

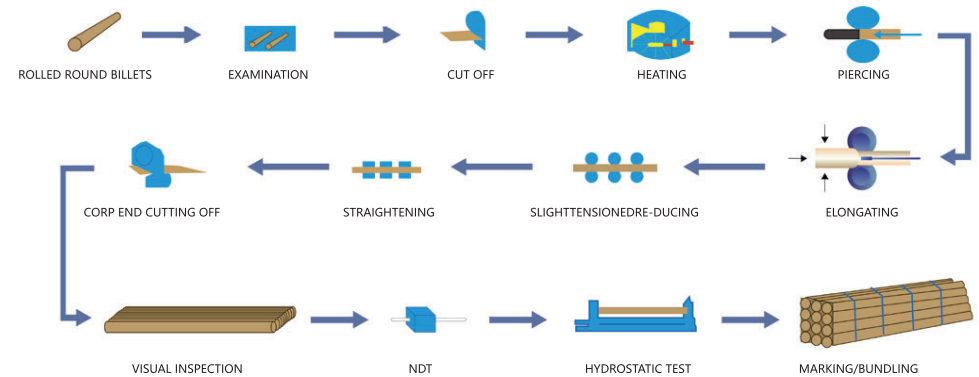
Cold drawn seamless tubing possesses a high strength-to-weight ratio and can be processed to provide a combination of desired characteristics.

Hot Rolling

Hot rolling is relative to cold rolling, cold rolling is rolling below the recrystallization temperature, and hot rolling is rolling above the recrystallization temperature.



HOT-ROLLING SEAMLESS STEEL PIPE PROCESS FLOW CHART



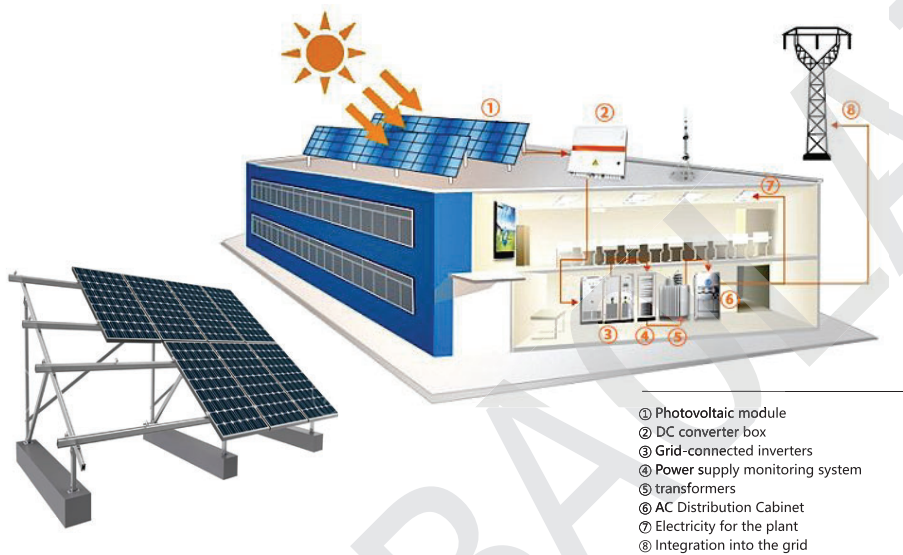
4 SOLAR MOUNTING SYSTEM

Baolai offers a wide range of solar mounting gear for all kinds of solar installations, including residential, industrial & commercial and utility scale projects. Focused on continuous innovation, Baolai's vision is to deliver the highest value to our clients at the lowest cost per kWh.

We also have the ability to customise mounting solutions to meet a client's specific requirements in a timely manner. We ship solar mounting structures world-wide and work through a network of strategic partners, distributors and agents.

Depending on its installation location, there are several types of mounting systems for PVs. For example, rooftop, ground mount, carports, post mount and even tracker solutions for residential or commercial applications.

Schematic diagram of distributed rooftop PV power plant



3 FIRE FIGHTING SYSTEM

Firefighting pipe is used to convey water or fire retardant. Usually it is carbon steel pipe in red FBE or galvanized coating. Grooved fittings are used for connecting Standpipe to control, distribute, or support pipeline in different sizes or directions. By groove connection, project time is saved a lot with fast installation and easy maintenance.

Baolai Steel provides most of fire sprinkler pipes and accessories in UL listed / FM approved.



ASTM A795 Black and Hot dipped Zinc-Coated, Welded Steel Pipe for Fire Protection Use

Available size for Sch10 pipe (Light-weight pipe)

Size			Thickness	Mass	Test pressure	Certificates
NPS	DN	OD(mm)	T(mm)	kg/m	MPa	
3/4"	20	26.7	2.11	1.28	4.8	ISO
1"	25	33.4	2.77	2.09	4.8	UL/FM
1-1/4"	32	42.2	2.77	2.69	6.9	UL/FM
1-1/2"	40	48.3	2.77	3.11	6.9	UL/FM
2"	50	60.3	2.77	3.93	6.9	UL/FM
2-1/2"	65	73	3.05	5.26	6.9	UL/FM
3"	80	88.9	3.05	6.46	6.9	UL/FM
4"	100	114.3	3.05	8.37	8.3	UL/FM
5"	125	141.3	3.4	11.56	8.3	UL/FM
6"	150	168.3	3.4	13.83	6.9	UL/FM
8"	200	219.1	4.78	25.26	5.5	UL/FM
10"	250	273.1	4.78	31.63	4.8	UL/FM

Note: For pipe NPS 8" and NPS 10", the thickness follows ASTM A795, not Sch10 of ASME B36.10M (8"/3.76mm, 10"/4.19mm).

Available size for Sch40 (STD) pipe

Size			Thickness	Mass	Test pressure	Certificates
NPS	DN	OD(mm)	T(mm)	kg/m	MPa	
1/2"	15	21.3	2.77	1.27	4.8	ISO
3/4"	20	26.7	2.87	1.69	4.8	ISO
1"	25	33.4	3.38	2.5	4.8	UL/FM
1-1/4"	32	42.2	3.56	3.39	6.9	UL/FM
1-1/2"	40	48.3	3.68	4.05	6.9	UL/FM
2"	50	60.3	3.91	5.44	6.9	UL/FM
2-1/2"	65	73	5.16	8.63	6.9	UL/FM
3"	80	88.9	5.49	11.29	6.9	UL/FM
4"	100	114.3	6.02	16.08	8.3	UL/FM
5"	125	141.3	6.55	21.77	8.3	UL/FM
6"	150	168.3	7.11	28.26	8.3	UL/FM
8"	200	219.1	7.04	36.82	8.3	UL/FM
10"	250	273.1	7.8	51.03	6.9	UL/FM

Note: For pipe NPS 8" and NPS 10", the thickness follows Sch30 of ASME B36.10M, not Sch40 of ASME B36.10M (8"/8.18mm, 10"/9.27mm).



ASTM A53 Pipe, Steel, Black And Hot-Dipped, Zinc-Coated, Welded And Seamless

Available size for Sch10 pipe (Light-weight pipe)

Size			Thickness	Mass	Test pressure	Certificates
NPS	DN	OD(mm)	T(mm)	kg/m	MPa	
1/2"	15	21.3	2.11	1	4.8	ISO
3/4"	20	26.7	2.11	1.28	4.8	ISO
1"	25	33.4	2.77	2.09	4.8	ISO
1-1/4"	32	42.2	2.77	2.69	9	ISO
1-1/2"	40	48.3	2.77	3.11	9	ISO
2"	50	60.3	2.77	3.93	13.2	ISO
2-1/2"	65	73	3.05	5.26	12	ISO
3"	80	88.9	3.05	6.46	9.9	ISO
4"	100	114.3	3.05	8.37	7.7	ISO
5"	125	141.3	3.4	11.56	6.9	ISO
6"	150	168.3	3.4	13.83	5.8	ISO
8"	200	219.1	3.76	19.97	4.9	ISO
10"	250	273	4.19	27.78	4.4	ISO

Available size for Sch40 (STD) pipe

Size			Thickness	Mass	Test pressure	Certificates
NPS	DN	OD(mm)	T(mm)	kg/m	MPa	
1/2"	15	21.3	2.77	1.27	4.8	ISO
3/4"	20	26.7	2.87	1.69	4.8	ISO
1"	25	33.4	3.38	2.50	4.8	UL/FM
1-1/4"	32	42.2	3.56	3.39	9	UL/FM
1-1/2"	40	48.3	3.68	4.05	9	UL/FM
2"	50	60.3	3.91	5.44	17.2	UL/FM
2-1/2"	65	73	5.16	8.63	17.2	UL/FM
3"	80	88.9	5.49	11.29	17.2	UL/FM
4"	100	114.3	6.02	16.08	15.2	UL/FM
5"	125	141.3	6.55	21.77	13.4	UL/FM
6"	150	168.3	7.11	28.26	12.3	UL/FM
8"	200	219.1	7.04	36.82	9.3	UL/FM
10"	250	273	7.80	51.01	8.3	UL/FM

Note: For pipe NPS 8" and NPS 10", the thickness follows Sch30 of ASME B36.10M, not Sch40 of ASME B36.10M (8" 7/8.18mm, 10" 9.27mm).

API 5L/ASTMA53 Welded Pipes/A106 Seamless Pipes

1) Dimensions, Weights, and Test Pressures for plain End Pipe

NPS Designator	DN Designator	Outside Diameter, in.[mm]	Nominal wall Thickness, in.[mm]	Nominal weight(Mass) per Unit Length, Plain End, lb/ft[kg/m]	Weight Class	Schedule No.	Test Pressure, psi [kPa] A53	
							Grade A	Grade B
1/2	15	0.840 [21.3]	0.109 [2.77]	0.85 [1.27]	STD	40	700 [4800]	700 [4800]
			0.147 [3.73]	1.09 [1.62]	XS	80	850 [5900]	850 [5900]
			0.188 [4.78]	1.31 [1.95]	-	160	900 [6200]	900 [6200]
			0.294 [7.47]	1.72 [2.55]	XXS	-	1000 [6900]	1000 [6900]
3/4	20	1.050 [26.7]	0.113 [2.87]	1.13 [1.69]	STD	40	700 [4800]	700 [4800]
			0.154 [3.91]	1.48 [2.20]	XS	80	850 [5900]	850 [5900]
			0.219 [5.56]	1.95 [2.90]	-	160	950 [6500]	950 [6500]
			0.308 [7.82]	2.44 [3.64]	XXS	-	1000 [6900]	1000 [6900]
1	25	1.315 [33.4]	0.113 [3.38]	1.68 [2.50]	STD	40	700 [4800]	700 [4800]
			0.179 [4.55]	2.17 [3.24]	XS	80	850 [5900]	850 [5900]
			0.250 [6.35]	2.85 [4.24]	-	160	950 [6500]	950 [6500]
			0.358 [9.09]	3.66 [5.45]	XXS	-	1000 [6900]	1000 [6900]
1 1/4	32	1.660 [42.2]	0.140 [3.56]	2.27 [3.39]	STD	40	1200 [8300]	1300 [9000]
			0.191 [4.85]	3.00 [4.47]	XS	80	1800 [12400]	1900 [13100]
			0.250 [6.35]	3.77 [5.61]	-	160	1900 [13100]	2000 [13800]
			0.382 [9.70]	5.22 [7.77]	XXS	-	2200 [15200]	2300 [15900]
1 1/2	40	1.900 [48.3]	0.145 [3.68]	2.72 [4.05]	STD	40	1200 [8300]	1300 [9000]
			0.200 [5.08]	3.63 [5.41]	XS	80	1800 [12400]	1900 [13100]
			0.281 [7.14]	4.86 [7.25]	-	160	1950 [13400]	2050 [14100]
			0.400 [10.16]	6.41 [9.56]	XXS	-	2200 [15200]	2300 [15900]
2	50	2.375 [60.3]	0.154 [3.91]	3.66 [5.44]	STD	40	2300 [15900]	2500 [17200]
			0.218 [5.54]	5.03 [7.48]	XS	80	2500 [17200]	2500 [17200]
			0.344 [8.74]	7.47 [11.11]	-	160	2500 [17200]	2500 [17200]
			0.436 [11.07]	9.04 [13.44]	XXS	-	2500 [17200]	2500 [17200]
2 1/2	65	2.875 [73.0]	0.203 [5.16]	5.80 [8.63]	STD	40	2500 [17200]	2500 [17200]
			0.276 [7.01]	7.67 [11.41]	XS	80	2500 [17200]	2500 [17200]
			0.375 [9.52]	10.02 [14.90]	-	160	2500 [17200]	2500 [17200]
			0.552 [14.02]	13.71 [20.39]	XXS	-	2500 [17 200]	2500 [17 200]
3	80	3.500 [88.9]	0.125 [3.18]	4.51 [6.72]	-	-	1290 [8900]	1500 [10300]
			0.156 [3.96]	5.58 [8.29]	-	-	1600 [11000]	1870 [12900]
			0.188 [4.78]	6.66 [9.92]	-	-	1930 [13300]	2260 [15600]
			0.216 [5.49]	7.58 [11.29]	STD	40	2220 [15300]	2500 [17200]
			0.250 [6.35]	8.69 [12.93]	-	-	2500 [17200]	2500 [17200]
			0.281 [7.14]	9.67 [14.40]	-	-	2500 [17200]	2500 [17200]
			0.300 [7.62]	10.26 [15.27]	XS	80	2500 [17200]	2500 [17200]
			0.438 [11.13]	14.34 [21.35]	-	160	2500 [17200]	2500 [17200]
			0.600 [15.24]	18.60 [27.68]	XXS	-	2500 [17 200]	2500 [17 200]
			0.125 [3.18]	5.18 [7.72]	-	-	1120 [7700]	1300 [19000]
			0.156 [3.96]	6.41 [9.53]	-	-	1400 [9700]	1640 [11300]
			0.188 [4.78]	7.66 [11.41]	-	-	1690 [11700]	1970 [13600]
3 1/2	90	4.000 [101.6]	0.226 [5.74]	9.12 [13.57]	STD	40	2030 [14000]	2370 [16300]
			0.250 [6.35]	10.02 [14.92]	-	-	2250 [15500]	2500 [17200]
			0.281 [7.14]	11.17 [16.63]	-	-	2500 [17200]	2500 [17200]
			0.318 [8.08]	12.52 [18.63]	XS	80	2800 [19300]	2800 [19300]
4	100	4.500 [114.3]	0.125 [3.18]	5.58 [8.71]	-	-	1000 [6900]	1170 [8100]
			0.156 [3.96]	7.24 [10.78]	-	-	1250 [8600]	1460 [10100]
			0.188 [4.78]	8.67 [12.91]	-	-	1500 [10300]	1750 [12100]
			0.219 [5.56]	10.02 [14.91]	-	-	1750 [12100]	2040 [14100]
			0.237 [6.02]	10.80 [16.07]	STD	40	1900 [13100]	2210 [15200]
			0.250 [6.35]	11.36 [16.90]	-	-	2000 [13800]	2330 [16100]
			0.281 [7.14]	12.67 [18.87]	-	-	2250 [15100]	2620 [18100]
			0.312 [7.92]	13.97 [20.78]	-	-	2500 [17200]	2800 [19300]
			0.337 [8.56]	15.00 [22.32]	XS	80	2700 [18 600]	2800 [19300]